

Algebra Exercises

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Chapter 8

Linear Algebra, Reprise

8.1 Functors

Exercise 8.1.1. Let $\mathcal{F} : \mathbf{C} \rightarrow \mathbf{D}$ be a covariant functor, and let $\mathbf{C}$ and $\mathbf{D}$ have products. Show there exists a unique morphism $\mathcal{F}(A \times B) \rightarrow \mathcal{F}(A) \times \mathcal{F}(B)$ making the relevant diagram commute.

If $\mathbf{D}$ has coproducts (denoted $A \amalg B$) and $\mathcal{G} : \mathbf{C} \rightarrow \mathbf{D}$ is contravariant, show there exists a unique morphism $\mathcal{G}(A) \amalg \mathcal{G}(B) \rightarrow \mathcal{G}(A \times B)$.

Solution. By definition of a functor, $\mathcal{F}(A \times B)$ is an object in $\mathbf{D}$, and

$$\begin{array}{ccc} & \mathcal{F}(A \times B) & \\ \mathcal{F}(\pi_A) \swarrow & & \searrow \mathcal{F}(\pi_B) \\ \mathcal{F}(A) & & \mathcal{F}(B) \end{array}$$

is a diagram in $\mathbf{D}$. Therefore, by universal property of products, the diagram

$$\begin{array}{ccccc} & & \mathcal{F}(\pi_A) & \xrightarrow{\quad} & \mathcal{F}(A) \\ & \nearrow & & \nearrow \pi_{\mathcal{F}(A)} & \\ \mathcal{F}(A \times B) & \dashrightarrow & \mathcal{F}(A) \times \mathcal{F}(B) & & \\ & \searrow & \searrow \pi_{\mathcal{F}(B)} & & \\ & & \mathcal{F}(B) & & \end{array}$$

$\mathcal{F}(\pi_B)$

commutes.

Now suppose $\mathcal{G} : \mathbf{C} \rightarrow \mathbf{D}$ is contravariant and $\mathbf{D}$ has coproducts. Since there exist projections $\pi_A : A \times B \rightarrow A$, $\pi_B : A \times B \rightarrow B$ in $\mathbf{C}$, and since $\mathcal{G}$ is contravariant, there exist morphisms

$$\mathcal{G}(\pi_A) : \mathcal{G}(A) \rightarrow \mathcal{G}(A \times B), \quad \mathcal{G}(\pi_B) : \mathcal{G}(B) \rightarrow \mathcal{G}(A \times B)$$

in $\mathbf{D}$. Since $\mathbf{D}$ has coproducts, by the universal property of coproducts,

$$\begin{array}{ccccc}
 \mathcal{G}(A) & & \xrightarrow{\mathcal{G}(\pi_A)} & & \mathcal{G}(A \times B) \\
 & \searrow i_A & & & \uparrow \\
 & \mathcal{G}(A) \amalg \mathcal{G}(B) & \dashrightarrow & & \mathcal{G}(A \times B) \\
 & \nearrow i_B & & & \uparrow \\
 \mathcal{G}(B) & & \xrightarrow{\mathcal{G}(\pi_B)} & & \mathcal{G}(A \times B)
 \end{array}$$

exists in $\mathbf{D}$ and commutes. ■

Exercise 8.1.2. Let $\mathcal{F} : \mathbf{C} \rightarrow \mathbf{D}$ be a fully faithful functor. If A, B are in $\mathbf{C}$, show that $A \cong B$ if and only if $\mathcal{F}(A) \cong \mathcal{F}(B)$.

Solution. Let $A \cong B$. Since $\mathcal{F}$ is a functor, commutative diagrams are sent to commutative diagrams, and thus

$$\begin{array}{ccccc}
 & & 1_A & & \\
 & \searrow & & \nearrow & \\
 A & \xrightarrow{\quad} & B & \xrightarrow{\quad} & A \\
 \downarrow \mathcal{F} & & & & \downarrow \mathcal{F} \\
 \mathcal{F}(A) & \longrightarrow & \mathcal{F}(B) & \longrightarrow & \mathcal{F}(A)
 \end{array}$$

commutes. It follows that the two morphisms in question under $\mathcal{F}$ still compose to $1_{\mathcal{F}(A)}$, and the same for B . Therefore, $\mathcal{F}(A) \cong \mathcal{F}(B)$.

Now the converse, let $\mathcal{F}(A) \cong \mathcal{F}(B)$. Since the morphism

$$\mathrm{Hom}_{\mathbf{C}}(A, B) \rightarrow \mathrm{Hom}_{\mathbf{D}}(\mathcal{F}(A), \mathcal{F}(B))$$

is a bijection, it follows that

$$\mathcal{F}(\alpha \circ \beta) = \mathcal{F}(1_{\mathcal{F}(A)}) \implies \alpha \circ \beta = \mathcal{F}^{-1}(1_{\mathcal{F}(A)}) = 1_A.$$

Therefore, $A \cong B$. ■

Exercise 8.1.3. Recall that a group G may be thought of as a groupoid $\mathbf{G}$ with a single object $\{*\}$. Prove that defining the action of a group G on an object A of a category $\mathbf{C}$ is the same as defining a functor $\mathbf{G} \rightarrow \mathbf{C}$.

Solution. Note that since $\mathbf{G}$ has one element, the action of a functor $\mathcal{F} : \mathbf{G} \rightarrow \mathbf{C}$ amounts to determining a single object A of $\mathbf{C}$ on which to map the single element. This is the same as choosing an object of $\mathbf{C}$ on which to act. Note also that a group action of G on A can be seen as determining an automorphism of A for all elements of G in a way compatible with the group operation (identity maps to identity, composition is the same as group multiplication). Since the

elements of G are viewed as morphisms in $\mathbf{G}$, and since functors preserve morphism composition and identities, we get that the structure of $\mathbf{G}$ is preserved for free. ■

Exercise 8.1.4. Let R be a commutative ring, and let $S \subseteq R$ be a *multiplicative subset* of R . Prove that localization is a functor; associating every R -module M with the localization $S^{-1}M$ and with every R -module homomorphism $\phi : M \rightarrow N$ the naturally induced homomorphism $S^{-1}M \rightarrow S^{-1}N$ defines a covariant functor $R\text{-Mod} \rightarrow S^{-1}R\text{-Mod}$.

Exercise 8.1.5. For F a field, denote by F^* the multiplicative group of nonzero elements of F . The assignment $\text{Fld} \rightarrow \text{Grp}$ mapping $F \mapsto F^*$ and a homomorphism of fields $\phi : k \rightarrow F$ to the restriction $\phi|_{k^*} : k^* \rightarrow F^*$ is clearly a covariant functor. We can also consider the category Fld_k^f of finite extensions of a fixed field k . Prove that the assignment $F \mapsto F^*$, together with the assignment of an extension $F_1 \subseteq F_2$ to the norm $N_{F_1 \subseteq F_2} : F_2^* \rightarrow F_1^*$ is a contravariant functor $\text{Fld}_k^f \rightarrow \text{Grp}$. State and prove an analogous statement for the trace.

Exercise 8.1.6. Formalize the notion of a presheaf of abelian groups on a topological space T . If $\mathcal{F}$ is a presheaf on T , elements of $\mathcal{F}(U)$ are called *sections* of $\mathcal{F}$ on U . The homomorphism $\rho_{UV} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$ induced by an inclusion $V \subseteq U$ is called the *restriction map*.

Show that a presheaf is obtained by letting $\mathcal{C}(U)$ be the additive abelian group of complex valued functions on U , with restriction of sections defined by ordinary restriction of functions.

For this presheaf, prove that one can uniquely glue sections agreeing on overlapping sets. That is, if U, V are open sets and $s_U \in \mathcal{C}(U)$, $s_V \in \mathcal{C}(V)$ agree after restriction to $U \cap V$, prove that there exists a unique $s \in \mathcal{C}(U \cup V)$ such that s restricts to s_U on U and s_V on V .

Solution. Formalizing the notion is very simple. I will change notation slightly: let $(X, \mathcal{T})$ be a topological space, and define the category $\mathbf{T}$ such that $\text{Obj}(\mathbf{T}) = \mathcal{T}$ and

$$\text{Hom}_{\mathbf{T}}(U, V) = \begin{cases} \emptyset, & U \not\subseteq V \\ \{*\}, & U \subseteq V \end{cases}.$$

Then a presheaf $\mathcal{F}$ on X is a contravariant functor $\mathcal{F} : \mathbf{T} \rightarrow \mathbf{Ab}$.

Let $\mathcal{C} : \mathbf{T} \rightarrow \mathbf{Ab}$ be the contravariant assignment given by

$$\mathcal{C}(U) = (C(U, \mathbb{C}), +).$$

That is, U maps to the set of continuous functions $f : U \rightarrow \mathbb{C}$, viewed as a group under addition. Morphisms $i : U \hookrightarrow V$ are defined as

$$\mathcal{C}(i)(f) = \rho_{VU}(f) = f|_U.$$

That is, the restriction to U of a section f of $\mathcal{C}$ over V is the standard function restriction. Verification that the group given is abelian is immediate, so now we show that the restriction morphisms are group homomorphisms. Note that if $f, g \in \mathcal{C}(V)$ and $U \subseteq V$, then

$$\rho_{VU}(f + g)(x) = (f + g)|_U(x) := f|_U(x) + g|_U(x) = \rho_{VU}(f)(x) + \rho_{VU}(g)(x).$$

Therefore, this is a group homomorphism by definition of the operation. Next, we show identity maps map to identity maps. Note that for any $f : U \rightarrow U$,

$$\rho_{UU}(f) = f_U = f.$$

Therefore, $\rho_{UU} = 1_{C(U)}$. Finally, we wish to show that if $U \subseteq V \subseteq W$, then $\rho_{WU} = \rho_{WV}\rho_{VU}$. Let $f : W \rightarrow \mathbb{C}$. We have

$$\rho_{WU}(f) = f_U, \quad \rho_{WV}\rho_{VU}(f) = \rho_{VU}f_V = f_U.$$

Therefore, $\mathcal{C}$ is a contravariant functor, and thus a presheaf of abelian groups on X .

Finally, the gluing condition. Let $s_U \in \mathcal{C}(U)$ and $s_V \in \mathcal{C}(V)$ agree on $U \cap V$. We can define a function $s \in \mathcal{C}(U \cup V)$ as the map

$$s(x) = \begin{cases} s_U(x), & x \in U \\ s_V(x), & x \in V \end{cases}.$$

Note that this is well-defined because if $x \in U$ and $x \in V$, then $s_U(x) = s_V(x)$. Moreover, it is manifestly unique since it is uniquely determined by s_U and s_V , and since they are continuous, s is clearly continuous as well due to the fact that s_U and s_V agree on $U \cap V$. Therefore, $\mathcal{C}$ is a sheaf of abelian groups on X . ■

Exercise 8.1.7. Define a topology on $\text{Spec } R$ by declaring the closed sets to be $V(I)$ for $I \subseteq R$ an ideal, and $V(I)$ the set of prime ideals containing I . This is the Zariski topology on $\text{Spec } R$.

- Verify that this is a topology on $\text{Spec } R$.
- Relate this to the Zariski topology as previously defined.
- Prove that $\text{Spec}(_)$ is a contravariant functor $\mathbf{CRing} \rightarrow \mathbf{Top}$.

Solution. Note that for any ring R , the ideal R is not prime. It follows that $V(R) = \emptyset$ and $V(0) = \text{Spec } R$. Now let $\{J_i\}_{i \in I}$ be a collection of ideals of R . Let

$$L := \bigcap_{i \in I} V(J_i).$$

Note that $\mathfrak{p} \in L$ if and only if $J_i \subseteq \mathfrak{p}$ for all $i \in I$. Define

$$K := \bigcup_{i \in I} J_i.$$

It follows that $(K) \subseteq \mathfrak{p}$, since (K) is the smallest ideal containing K . Also note that if $\mathfrak{p} \in V((K))$, then $J_i \subseteq K \subseteq (K) \subseteq \mathfrak{p}$ for all i , and thus $\mathfrak{p} \in V(L)$. Therefore, $V(L) = V((K))$. Therefore, closed sets are closed under intersection (and thus open sets are closed under union). ■

Exercise 8.1.8. K alg closed field, something with $K\text{-Aff}$ and $\mathbb{A}_K^0$.

Exercise 8.1.9. Let $\mathbf{C}$, $\mathbf{D}$ be categories, and assume $\mathbf{C}$ to be small. Define a functor category $\mathbf{D}^{\mathbf{C}}$, whos objects are covariant functors $\mathbf{C} \rightarrow \mathbf{D}$ and whos morphisms are natural transformations.

Prove that the assignment $X \mapsto h_X$ defines a covariant functor $\mathbf{C} \rightarrow \mathbf{Set}^{\mathbf{C}^{op}}$.

Solution. The construction $\mathbf{D}^{\mathbf{C}}$ is clearly a category. The natural transformation $1_{\mathcal{F}} : \mathcal{F} \Rightarrow \mathcal{F}$ can be defined as the collection of morphisms $\mathcal{F}(1_X) = 1_{\mathcal{F}(X)}$ for X in $\mathbf{C}$. Let $\mathcal{F}, \mathcal{G}, \mathcal{H} \in \text{Obj}(\mathbf{D}^{\mathbf{C}})$, and let $\eta : \mathcal{F} \Rightarrow \mathcal{G}$, $\nu : \mathcal{G} \Rightarrow \mathcal{H}$. Note that by definition of a natural transformation, the two squares

$$\begin{array}{ccccc} \mathcal{F}(X) & \xrightarrow{\eta_X} & \mathcal{G}(X) & \xrightarrow{\nu_X} & \mathcal{H}(X) \\ \downarrow \mathcal{F}(\alpha) & & \downarrow \mathcal{G}(\alpha) & & \downarrow \mathcal{H}(\alpha) \\ \mathcal{F}(Y) & \xrightarrow{\eta_Y} & \mathcal{G}(Y) & \xrightarrow{\nu_Y} & \mathcal{H}(Y) \end{array}$$

commute. Therefore, we have a natural transformation $\nu\eta : \mathcal{F} \Rightarrow \mathcal{H}$ defined in this way.

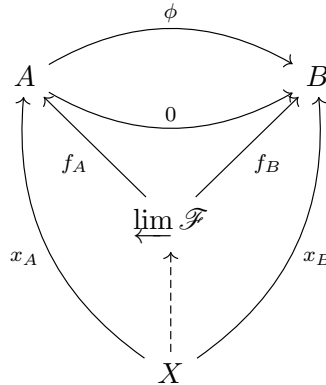
Defining morphisms in the second part is so hard wtf ■

Exercise 8.1.10. Let $\mathbf{Z}$ be the empty category consisting of no objects and no morphisms. One can contemplate a functor $\mathcal{Z} : \mathbf{Z} \rightarrow \mathbf{C}$. What is $\varprojlim \mathcal{Z}$, if such a category exists?

Solution. Note that a cone in this category is simply an object L , since there need not exist morphisms $\lambda_I : L \rightarrow \mathcal{Z}(I)$ since $\mathcal{Z}(I)$ does not exist. Thus, an object that is final in the category of cones is simply an object that is final in the category $\mathbf{C}$, and thus a limit is simply a final object. ■

Exercise 8.1.11. Verify that in $R\text{-Mod}$, the equalizer of $\phi : A \rightarrow B$ and $0 : A \rightarrow B$ is the kernel $\ker \phi$.

Solution. Verification is simple. Note that for all objects X , the diagram



commutes. Moreover, $f_B = 0f_A = \phi f_A$. However, since $0f_A = 0$, we have $\phi f_A = 0$, and $f_B = 0$.

We also have $x_B = 0x_A = \phi x_A$, and thus $x_B = 0$. We can rewrite this diagram as

$$\begin{array}{ccccc}
 & & 0 & & \\
 & \curvearrowright & & \curvearrowleft & \\
 X & \xrightarrow{x_A} & A & \xrightarrow{\phi} & B \\
 & \searrow \text{dashed} & \uparrow f_A & \nearrow 0 & \\
 & & \varprojlim \mathcal{F} & &
 \end{array}$$

Note that this is precisely the universal property satisfied by the kernel, since by commutativity of the diagram, we have that $\phi f_A = 0$, and we can actually drop the 0 morphism $\varprojlim \mathcal{F} \rightarrow B$ while retaining precisely the same information. Therefore, the same proof given that kernels satisfy this universal property in $R\text{-Mod}$ suffices here to show that $\varprojlim \mathcal{F}$ exists, and is the kernel. ■

Exercise 8.1.12. Let $\mathbf{I}$ be the category consisting of elements of $\mathbb{Z}^+$ as objects, and a unique morphism $i \rightarrow j$ if and only if $i \leq j$. Let $\mathcal{F} : \mathbf{I} \rightarrow R\text{-Mod}$ be a functor, consisting of a choice of object A_i for each i , and a morphism $\phi_{ji} : A_i \rightarrow A_j$ for all i, j . Note that an element of

$$\prod_i A_i$$

is a sequence $(a_i)_{i>0}$. Call a sequence (a_i) coherent if $\phi_{i+1,i}(a_{i+1}) = a_i$ for all i . Let A be the submodule of $\prod_i A_i$ consisting of all coherent sequences. Let $\phi_i : A \rightarrow A_i$ be the restriction of the canonical projection $\pi_i : \prod_i A_i \rightarrow A_i$ to A . Show that A , together with the morphisms ϕ_i , satisfy

$$\varprojlim_i A_i = A.$$

Solution. First, I want to show A is truly an R -module. Let $(a_i), (b_i) \in A$. Then

$$(a_i) - (b_i) = (a_i - b_i).$$

Note that (with the mild abuse of notation), we have

$$\phi_{i+1,i}(a_{i+1} - b_{i+1}) = \phi_{i+1,i}(a_{i+1}) - \phi_{i+1,i}(b_{i+1}) = (a_i) - (b_i) = (a_i - b_i).$$

Additionally, for $r \in R$,

$$r(a_i) = (ra_i).$$

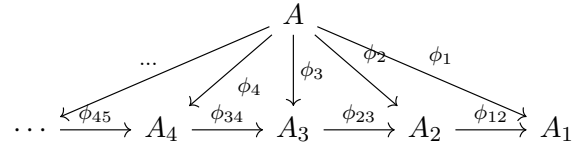
Thus,

$$\phi_{i+1,i}(ra_{i+1}) = r\phi_{i+1,i}(a_{i+1}) = r(a_i) = (ra_i).$$

Therefore, A is an R -module, as perscribed. Now we show the relevant diagram commutes. Let $n \in \mathbb{Z}^+$. We have $\phi_n(a_i) = a_n$. We also have

$$(\phi_{n+1,n}\phi_{n+1}) (a_i) = \phi_{n+1,n}(a_{n+1}) = a_n.$$

Therefore, the diagram



clearly commutes. Finally, A is manifestly final due to being the subset of a product and inheriting the canonical projections. ■